Agenda

- ➔ KNN Algorithm
- ➔ Sklearn KNN
- ➔ SVM Algorithm
- ➔ Sklearn SVM
- ➔ First Classification Project

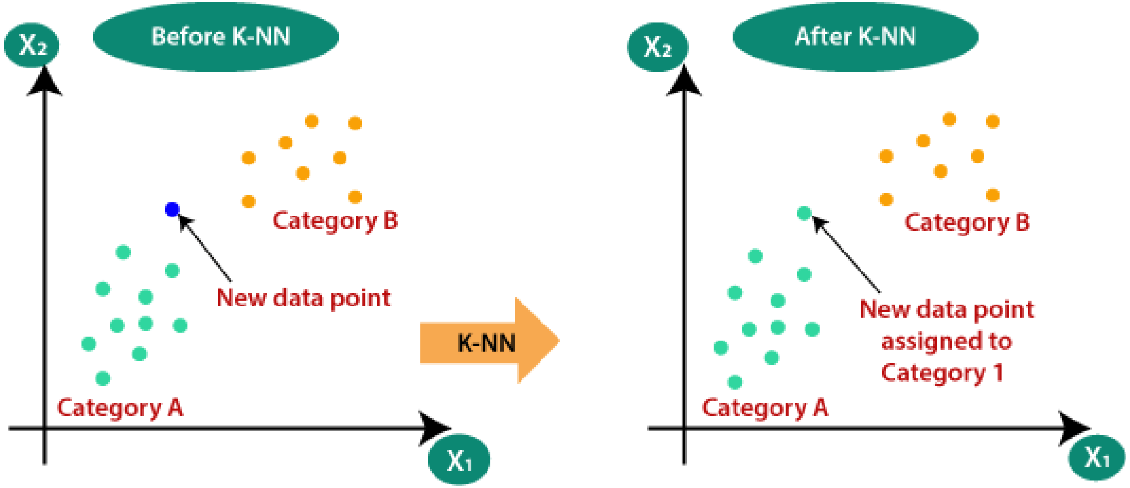
1. KNN Algorithm

K-Nearest Neighbour Algorithm:

- ➔ K-Nearest Neighbour is one of the simplest Machine Learning algorithms based on Supervised Learning technique.
- ➔ K-NN algorithm assumes the similarity between the new case/data and available cases and put the new case into the category that is most similar to the available categories.
- ➔ K-NN algorithm stores all the available data and classifies a new data point based on the similarity. This means when new data appears then it can be easily classified into a well suite category by using K- NN algorithm.

K-Nearest Neighbour Algorithm:

- Suppose there are two classes, i.e., Class A and Class B, and we need to classify a new data point to one of these classes.
- To solve this type of problem, we need a K-NN algorithm. So that, we can easily identify the category or class of a particular dataset.

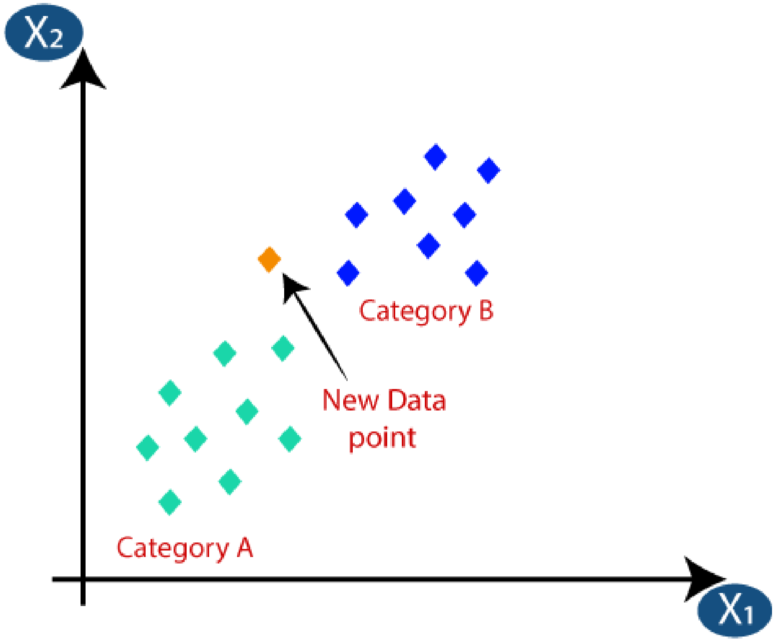


How does K-NN work?

- ➔ Step-1: Select the number K of the neighbours.
- ➔ Step-2: Calculate the Euclidean distance of all neighbours.
- ➔ Step-3: Take the K nearest neighbours as per the calculated Euclidean distance.
- ➔ Step-4: Among these k neighbours, count the number of the data points in each category.
- ➔ Step-5: Assign the new data point to that category for which the number of the neighbour is maximum.
- ➔ Step-6: Our model is ready.

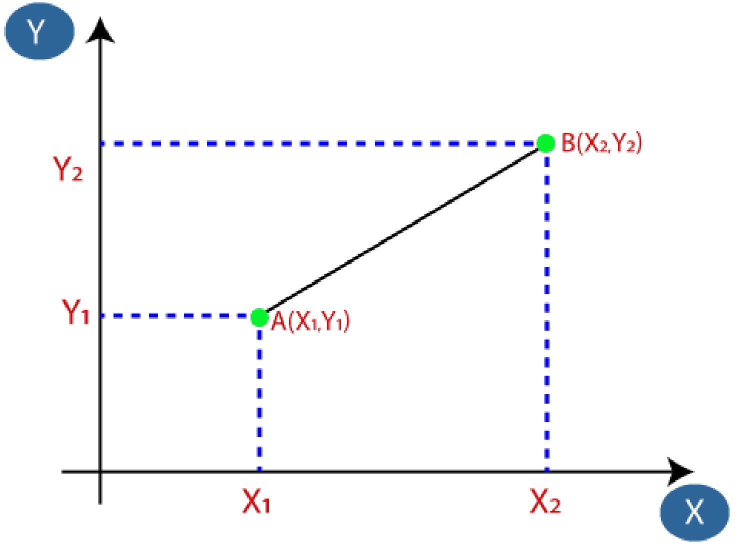
Example:

➔ Suppose we have a new data point and we need to put it in the required class.



Example:

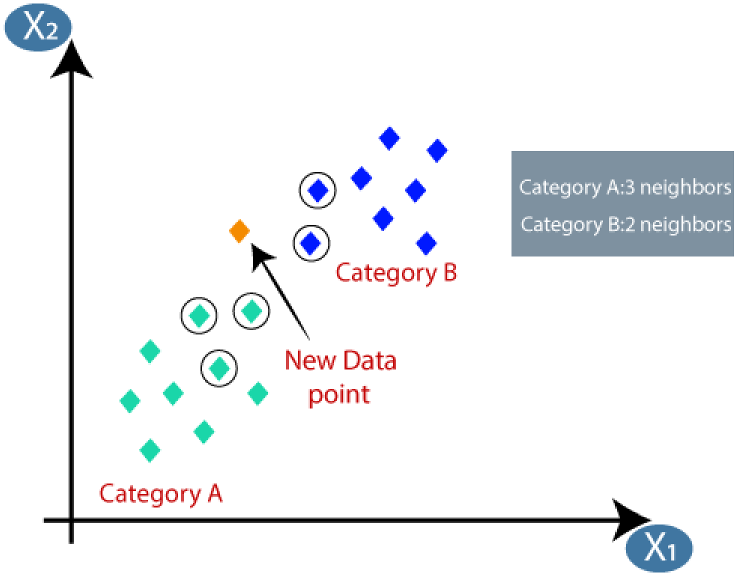
- ➔ Firstly, we choose the number of neighbours, k=5.
- ➔ Next, we calculate the Euclidean distance between all neighbours and the new data point.



Euclidean Distance between A_1 and B_2 = $\sqrt{(X_2-X_1)^2+(Y_2-Y_1)^2}$

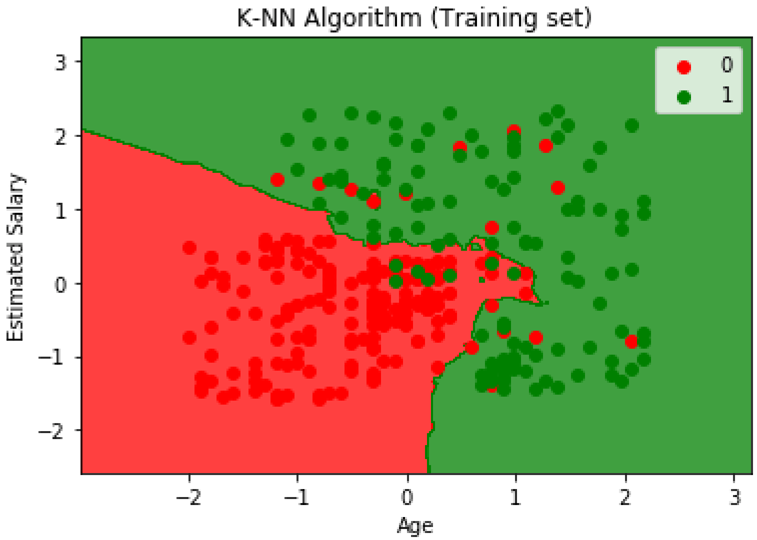
Example:

- ➔ By calculating the Euclidean distance we got the nearest K neighbours, as three neighbours in category A and two neighbours in category B.
- ➔ Hence this new data point must belong to category A

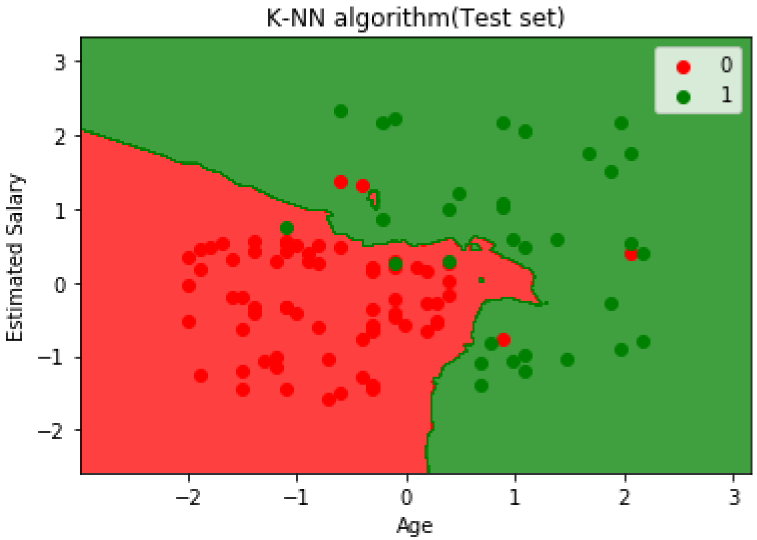


2. Sklearn KNN

Sklearn KNN:



K-NN Algorithm (Training set)
Estimated Salary
Age
0
1
3
2
1
0
−1
−2
−2
−1
0
1
2
3

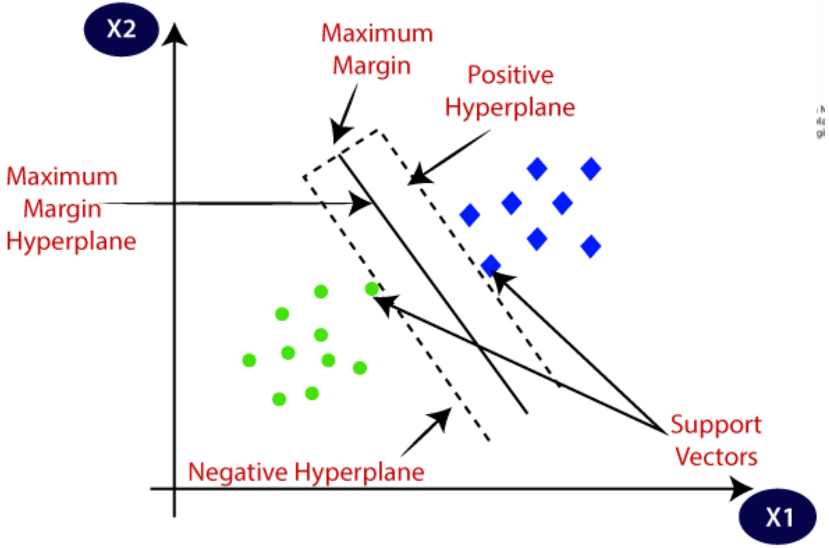


K-NN algorithm(Test set)
Estimated Salary
Age
0
1
3
2
1
0
−1
−2
−2
−1
0
1
2
3

3. SVM Algorithm

Support Vector Machine Algorithm:

- ➔ The goal of the SVM algorithm is to create the best decision boundary that can segregate n-dimensional space into classes. This best decision boundary is called a **Hyperplane**.
- ➔ SVM chooses the extreme points/vectors that help in creating the Hyperplane. These extreme cases are called as **Support Vectors**.



Types of SVM:

Linear SVM:

➔ Linear SVM is used for linearly separable data, which means if a dataset can be classified into two classes by using a single straight line, then such data is termed as linearly separable data, and classifier is used called as Linear SVM classifier.

Non-linear SVM:

➔ Non-Linear SVM is used for non-linearly separated data, which means if a dataset cannot be classified by using a straight line, then such data is termed as non-linear data and classifier used is called as Non-linear SVM classifier.

Kernel Functions:

- Kernel Function is a method used to take data as input and transform into the required form of processing data.
- Kernel Function generally transforms the training set of data so that a non-linear decision surface is able to be transformed to a linear equation in a higher number of dimension spaces.
- The kernel function is what is applied on each data instance to map the original non-linear observations into a higher-dimensional space in which they become separable.
- Standard Kernel Function Equation:

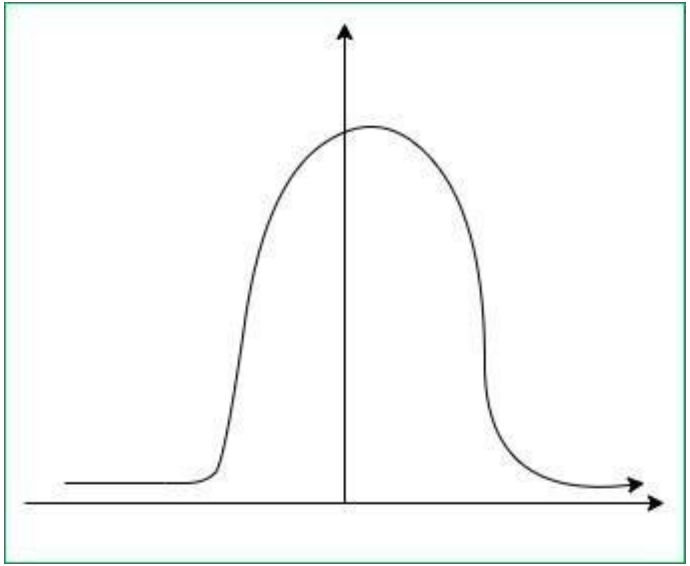
$$K(\bar{x}) = 1, if ||\bar{x}|| <= 1$$

$$K(\bar{x}) = 0, Otherwise$$

Kernel Functions:

➔ Gaussian Kernel Radial Basis Function (RBF):

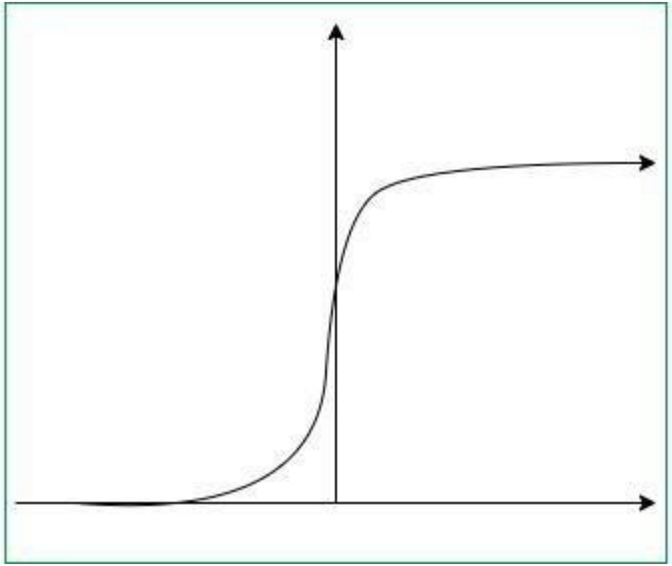
$$K(x, y) = e^{-}(\gamma||x - y||^2)$$



Kernel Functions:

➔ Sigmoid Kernel:

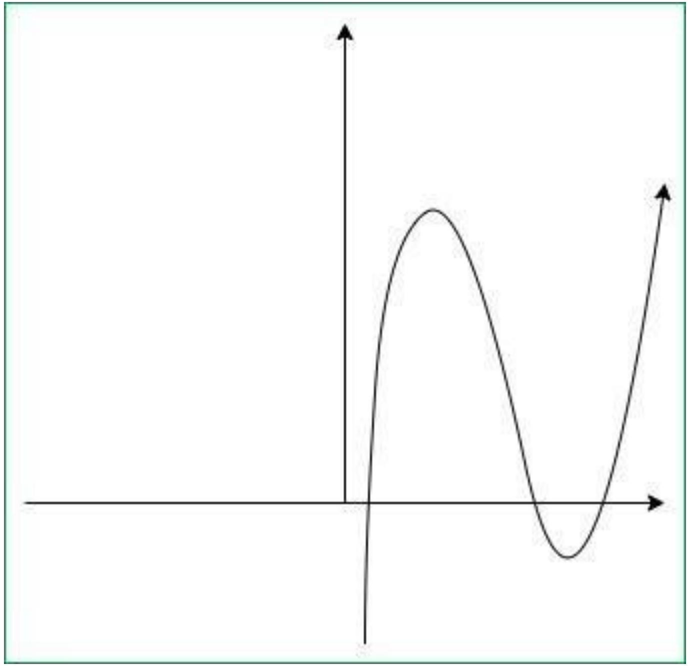
$$K(x, y) = tanh(\gamma . x^T y + r)$$



Kernel Functions:

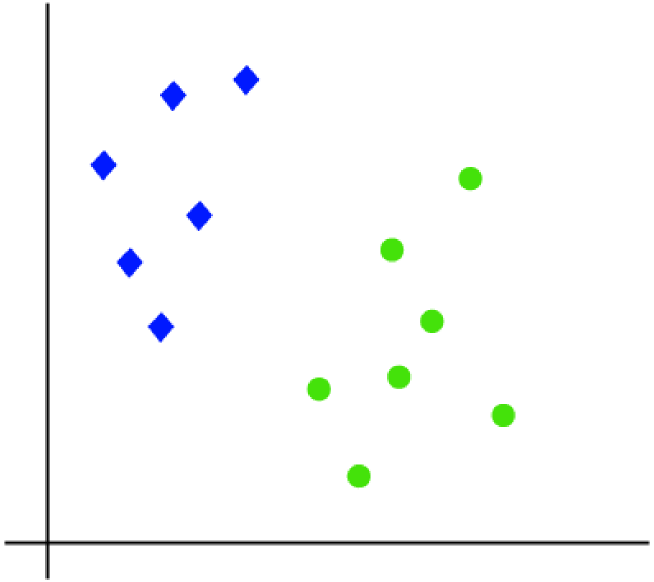
➔ Polynomial Kernel:

$$K(x, y) = tanh(\gamma . x^T y + r)^d, \gamma > 0$$



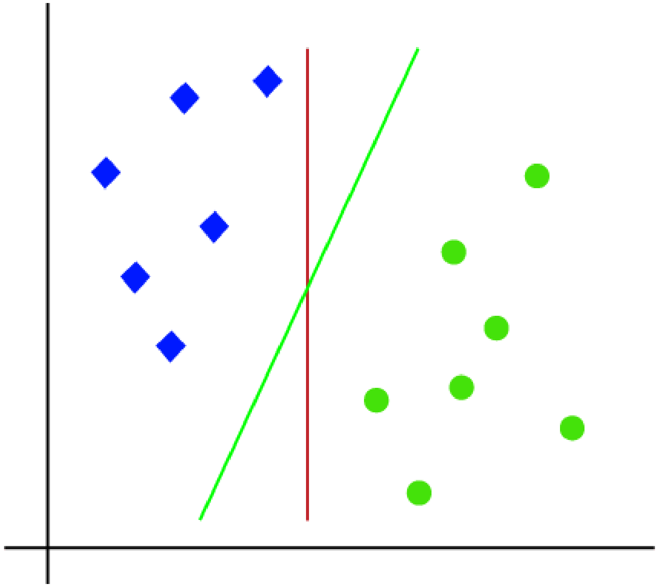
Example "Linear SVM":

➔ Suppose we have a dataset that has two classes (green and blue), and the dataset has two features x1 and x2. We want a classifier that can classify the pair(x1, x2) of coordinates in either green or blue.



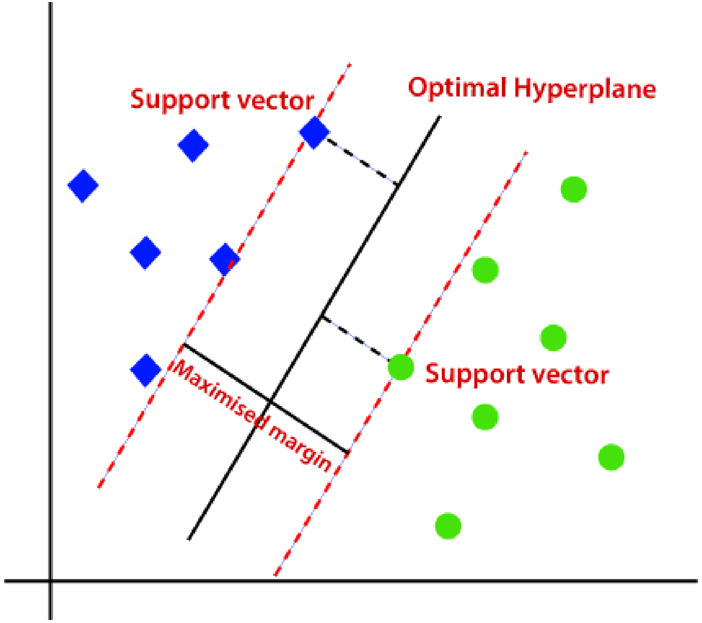
Example “Linear SVM”:

- ➔ So as it is 2-d space so by just using a straight line, we can easily separate these two classes. But there can be multiple lines that can separate these classes.



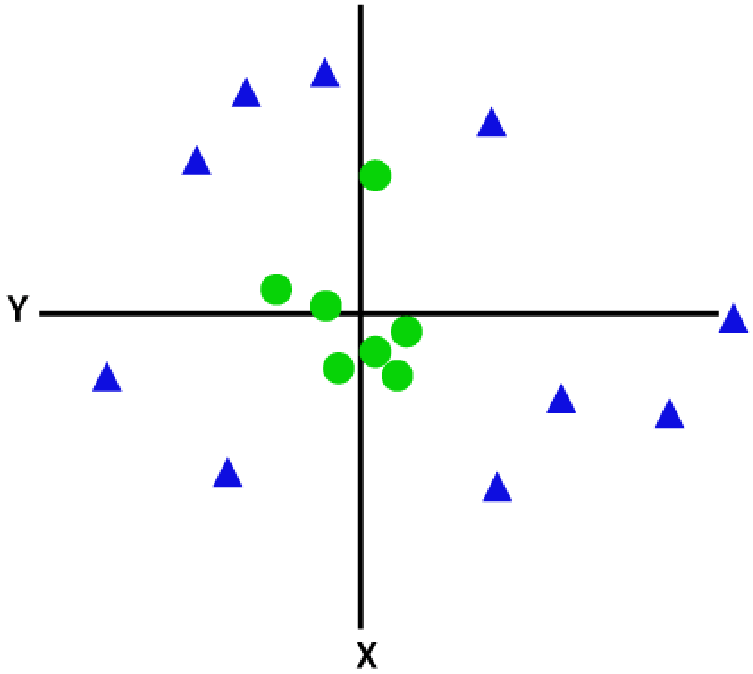
Example “Linear SVM”:

- ➔ To find the Hyperplane, SVM finds the closest point of the line from both the classes “Support Vectors”.
- ➔ The distance between the vectors and the hyperplane is called as Margin.
- ➔ The goal of SVM is to maximize this margin.
- ➔ The hyperplane with maximum margin is called the optimal hyperplane.



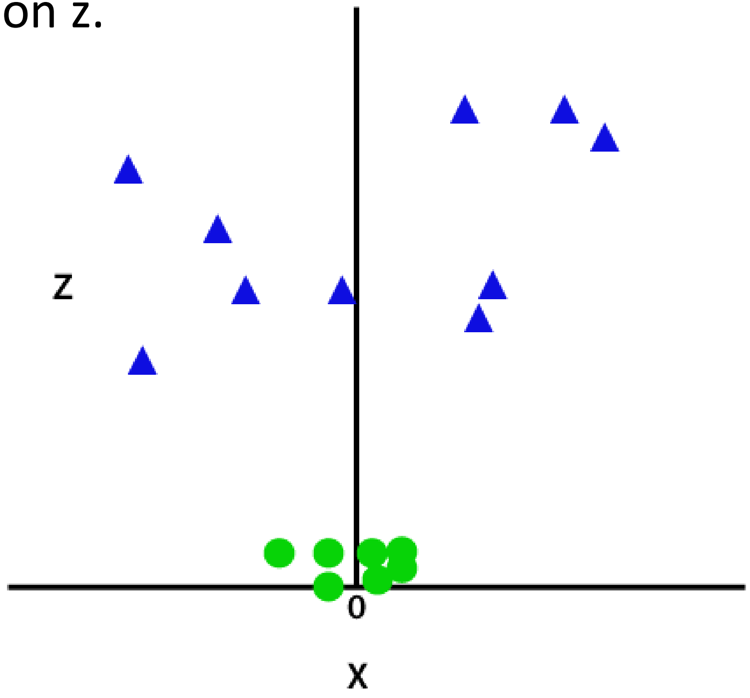
Example "Non-Linear SVM":

➔ For non-linear data, we cannot draw a single straight line.



Example "Non-Linear SVM":

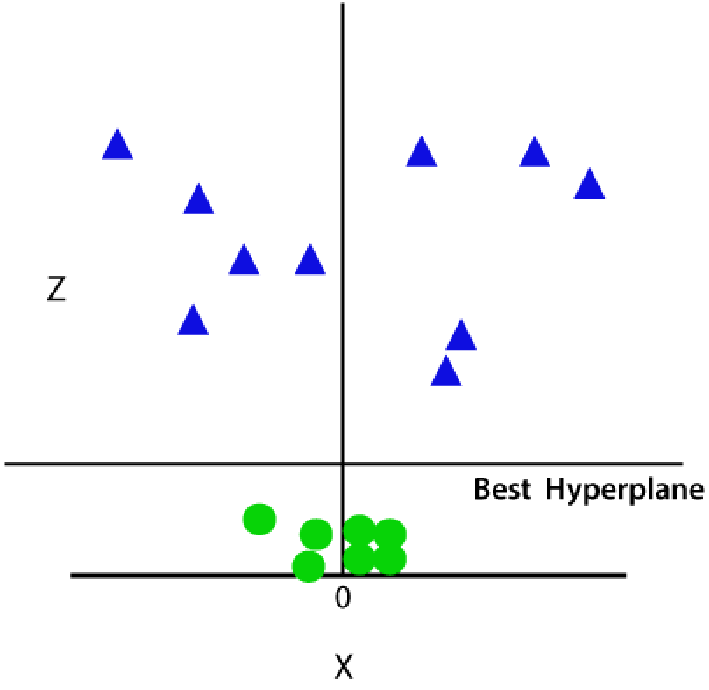
- ➔ To separate these data points, we need to add one more dimension.
- ➔ We will add a third dimension z.
- ➔ $z = x^2 + y^2$



Example “Non-Linear SVM”:

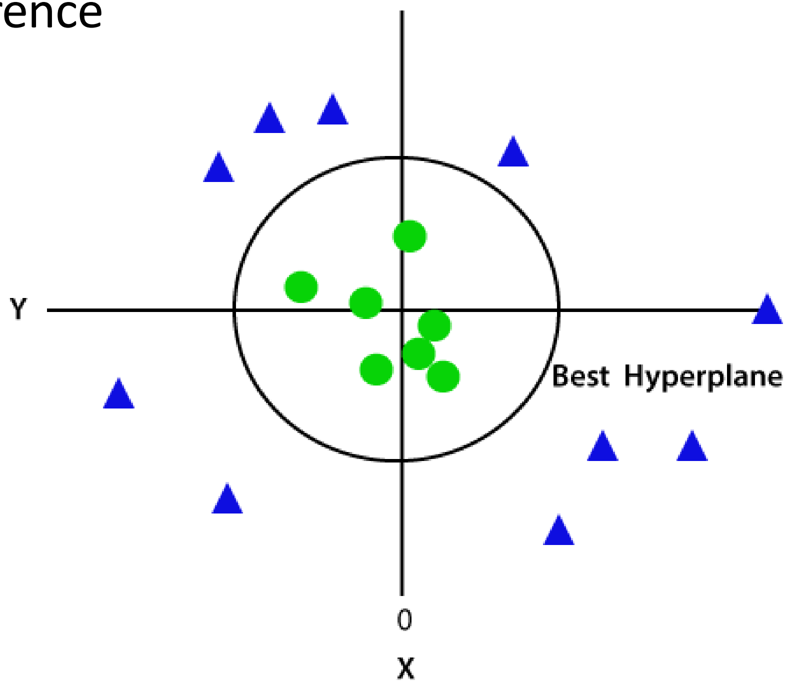
➔ Now, SVM will divide the datasets into classes in the following way.

➔ Since we are in 3-d Space, hence it is looking like a plane parallel to the x-axis.



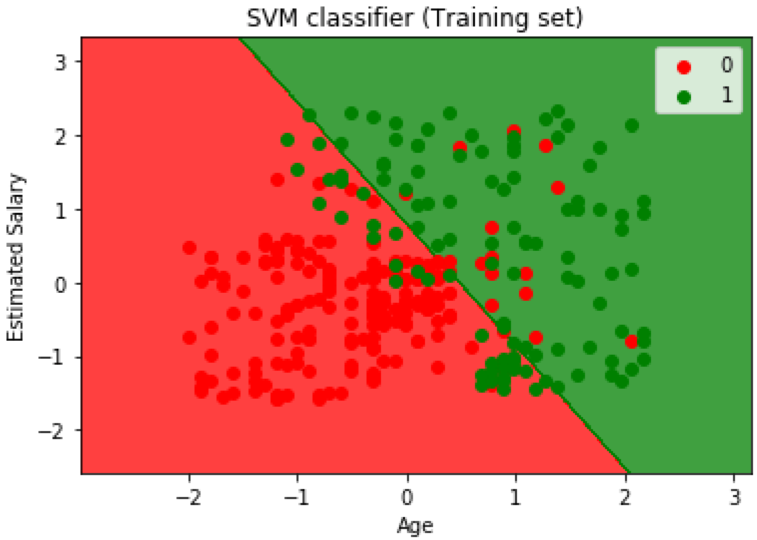
Example “Non-Linear SVM”:

- ➔ If we convert it in 2d space with z=1, then it will become as below.
- ➔ Hence we get a circumference of radius 1 in case of non-linear data.

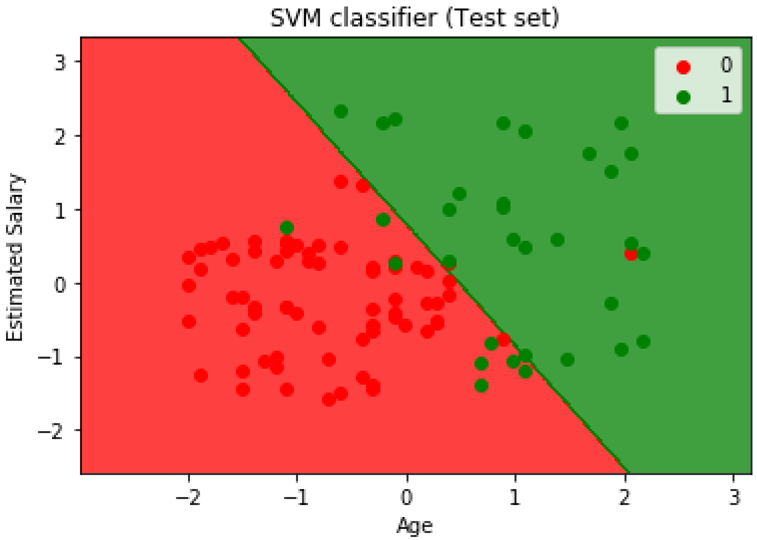


4. Sklearn SVM

Sklearn SVM:



SVM classifier (Training set)
0
1
Estimated Salary
3
2
1
0
−1
−2
−2
−1
0
1
2
3
Age



SVM classifier (Test set)
0
1
Estimated Salary
3
2
1
0
−1
−2
−2
−1
0
1
2
3
Age

5. First Classification Project

Classification Project:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
5	6	0	3	Moran, Mr. James	male	NaN	0	0	330877	8.4583	NaN	Q
6	7	0	1	McCarthy, Mr. Timothy J	male	54.0	0	0	17463	51.8625	E46	S
7	8	0	3	Palsson, Master. Gosta Leonard	male	2.0	3	1	349909	21.0750	NaN	S
8	9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27.0	0	2	347742	11.1333	NaN	S
9	10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14.0	1	0	237736	30.0708	NaN	C

Any Questions?